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Inventor(s)	Takahashi; Yuki

Vehicle image display system, vehicle image display method, and storage medium

Abstract

A vehicle image display system includes: a display device for displaying a digital image which is an image based on image data, a dark condition determination unit for determining whether a dark condition is satisfied with respect to the brightness of a surrounding environment of the vehicle, a luminance determination unit for determining whether the image data includes a high luminance pixel, a pixel value determination unit for determining whether a pixel having a pixel value equal to or lower than a second threshold value is not included in a peripheral pixel, and an image adjustment unit for performing a first brightness adjustment process for reducing the brightness of the image data when the dark condition is satisfied and the image data includes a high luminance pixel and the peripheral pixel does not include a pixel having a pixel value equal to or lower than the second threshold value.

Inventors:	Takahashi; Yuki (Nagoya, JP)
Applicant:	TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota, JP)
Family ID:	1000008782004
Assignee:	TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota, JP)
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Japanese Patent Application No. 2022-164313 filed on Oct. 12, 2022 incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to a vehicle image display system, a vehicle image display method, and a storage medium.

2. Description of Related Art

(3) In Japanese Unexamined Patent Application Publication No. 2009-035162 (JP 2009-035162 A) described below, when a luminance value of a part of image data (captured image) acquired by a camera is equal to or larger than a predetermined value, image processing for reducing the brightness of the part is performed. In this case, the luminance value of the remaining portion excluding the part of the image data is smaller than the predetermined value. In JP 2009-035162 A, image processing for adjusting the exposure time and the amount of received light of the image data is performed based on the luminance information of the remaining portion. Further, in JP 2009-035162 A, a display device displays the image data subjected to such image processing.

SUMMARY

(4) Here, it is assumed that the above predetermined value is set to a large value. In this case, the brightness of the image data is not reduced when the light emitted by a lighting device of another vehicle directly hits the camera at night. Therefore, the image based on the image data displayed on the display device is blurred. The driver who sees the display device may feel glare. On the other hand, when the above predetermined value is set to a small value, the brightness of image data representing a white subject such as a white line on a road may be reduced more than necessary.

(5) In view of the above, it is an object of the present disclosure to provide a vehicle image display system, a vehicle image display method, and a storage medium that suppress a person who sees a display device that displays a digital image generated by an imaging device from feeling glare when a strong light enters the imaging device and that do not reduce the brightness of the digital image more than necessary.

(6) A vehicle image display system according to claim 1 includes: an imaging device that images a subject located around a vehicle; a display device that displays a digital image that is an image based on image data generated by the imaging device; a dark condition determination unit that determines whether a dark condition is satisfied with respect to a brightness of a surrounding environment of the vehicle; a luminance determination unit that determines whether a high luminance pixel that is a pixel of a luminance equal to or larger than a first threshold value is included in the image data; a pixel value determination unit that determines whether a pixel having a pixel value equal to or smaller than a second threshold value is not included in a peripheral pixel that is a pixel around the high luminance pixel; and an image adjustment unit that performs a first brightness adjustment process for reducing a brightness of the image data, when the dark condition is satisfied and a processing condition is satisfied, the processing condition being a condition that is satisfied when the image data includes the high luminance pixel and the peripheral pixel does not include a pixel having a pixel value equal to or smaller than the second threshold value.

(7) In the vehicle image display system according to claim 1, the processing condition is satisfied when the dark condition determination unit determines that the dark condition is satisfied, the luminance determination unit determines that the image data includes a high luminance pixel that is

a pixel having a luminance equal to or larger than the first threshold value, and the pixel value determination unit determines that the peripheral pixel that is a pixel around the high luminance pixel does not include a pixel having a pixel value equal to or smaller than the second threshold value. Further, when the processing condition is satisfied, the image adjustment unit performs the first brightness adjustment process for reducing the brightness of the image data. When a high luminance pixel is included in the image data and a pixel having a pixel value equal to or smaller than the second threshold value is not included in the peripheral pixel due to the strong light entering the imaging device as described above, so-called white blur may have occurred in the peripheral pixel. Therefore, in this case, the first brightness adjustment process for reducing the brightness is performed on the image data. Accordingly, when a strong light enters the imaging device, it is less likely for a person who sees the display device displaying a digital image, which is an image based on the image data, to feel glare.

(8) When the dark condition is not satisfied, the image data does not include a high luminance pixel, or the peripheral pixel includes a pixel having a pixel value equal to or smaller than the second threshold value, there is a high possibility that no white blur has occurred in the digital image based on the peripheral pixel. Therefore, the vehicle image display system according to claim **1** does not perform the first brightness adjustment process on the image data in this case. As a result, the vehicle image display system according to claim **1** does not reduce the brightness of the image data more than necessary in such a case.

(9) In the vehicle image display system according to claim **2**, when the dark condition is satisfied and the processing condition is not satisfied, the image adjustment unit performs a second brightness adjustment process different from the first brightness adjustment process on the image data.

(10) In the vehicle image display system according to claim **2**, when the dark condition is satisfied and the processing condition is not satisfied, the image adjustment unit performs a second brightness adjustment process different from the first brightness adjustment process on the image data. Therefore, the vehicle image display system according to claim **2** can adjust the brightness of the image data so that the occupant of the vehicle can easily visually recognize the digital image displayed on the display device when the processing condition is not satisfied.

(11) In the vehicle image display system according to claim **3**, the dark condition is satisfied when an average luminance of the image data is equal to or smaller than a third threshold value or when an illuminance around the vehicle is equal to or smaller than a fourth threshold value.

(12) The dark condition determination unit of the vehicle image display system according to claim **3** can accurately determine whether the surrounding environment of the vehicle is dark. Therefore, the vehicle image display system according to claim **3** can accurately determine whether there is a risk of occurrence of white blur.

(13) A vehicle image display method according to claim **4** includes: a step of determining whether a dark condition is satisfied with respect to a brightness of a surrounding environment of a vehicle; a step of determining whether a high luminance pixel that is a pixel of a luminance equal to or larger than a first threshold value is included in an image data generated by an imaging device that images a subject located around the vehicle; a step of determining whether a pixel having a pixel value equal to or smaller than a second threshold value is not included in a peripheral pixel that is a pixel around the high luminance pixel; a step of performing a first brightness adjustment process for reducing a brightness of the image data, when the dark condition is satisfied, the image data includes the high luminance pixel, and the peripheral pixel does not include a pixel having a pixel value equal to or smaller than the second threshold value; and a step of displaying a digital image that is an image based on the image data subjected to the first brightness adjustment process.

(14) A storage medium according to claim **5** stores a program causing a computer to execute processes including: a process of determining whether a dark condition is satisfied with respect to a brightness of a surrounding environment of a vehicle; a process of determining whether a high

luminance pixel that is a pixel of a luminance equal to or larger than a first threshold value is included in an image data generated by an imaging device that images a subject located around the vehicle; a process of determining whether a pixel having a pixel value equal to or smaller than a second threshold value is not included in a peripheral pixel that is a pixel around the high luminance pixel; a process of performing a first brightness adjustment process for reducing a brightness of the image data, when the dark condition is satisfied, the image data includes the high luminance pixel, and the peripheral pixel does not include a pixel having a pixel value equal to or smaller than the second threshold value; and a process of displaying a digital image that is an image based on the image data subjected to the first brightness adjustment process.

(15) As described above, the vehicle image display system, the vehicle image display method, and the storage medium according to the present disclosure have an excellent effect that a person who sees a display device that displays a digital image generated by an imaging device is suppressed from feeling glare when a strong light enters the imaging device and that the brightness of the digital image is not reduced more than necessary.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

(2) FIG. 1 is a side view of a vehicle and a subsequent vehicle on which a vehicle image display system according to an embodiment is mounted;

(3) FIG. 2 is a control diagram of an ECU mounted on vehicles;

(4) FIG. 3 is a functional diagram of an ECU;

(5) FIG. 4 is a flow chart illustrating a process executed by CPU;

(6) FIG. 5 is a diagram showing a display device of a rearview mirror displaying digital images in which white blur has occurred; and

(7) FIG. 6 is a diagram illustrating a display device of a rearview mirror displaying a digital image in which white blur does not occur.

DETAILED DESCRIPTION OF EMBODIMENTS

(8) Hereinafter, a vehicle image display system, a vehicle image display method, and a program according to the present disclosure will be described with reference to the drawings.

(9) The vehicle **10** shown in FIG. 1 is equipped with a vehicle image display system **25** (hereinafter, system **25**). The vehicle **10** includes a vehicle body **11**, a camera (imaging device) **14**, a rearview mirror **16**, and an ECU **20**. It should be noted that the vehicles **10** can execute driving assistance control of the level **1** to the level **5** determined by Society of Automotive Engineers (SAE) (American Automotive Engineering Association). Further, the brightness adjustment processing described below can be executed while the vehicle **10** executes the driving assistance control and during the execution of the manual driving.

(10) A camera **14** is provided at a rear portion of the vehicle body **11**. The camera **14** includes an image sensor and a lens (both not shown). The image sensor includes a pixel array unit having a plurality of pixels (dots). Each pixel outputs the light incident on the pixel array unit via the lens as an electric signal. Further, the image sensor includes an exposure adjustment system (electronic shutter).

(11) The camera **14** is capable of capturing an image of a subject located behind the vehicle **10**. For example, as shown in FIG. 1, when the vehicle **10** is traveling forward on the road **40**, the camera **14** is capable of capturing an image of a succeeding vehicle (subject) **30** traveling forward on the road **40** behind the vehicle **10**. A pair of left and right lighting devices **31** (only the lighting device

31 on the left side is shown in FIG. 1) is provided at the front end portion of the following vehicle **30**.

(12) A rearview mirror **16** is provided inside the vehicle **10**. A digital display device **17** that displays a digital image based on the image data generated by the camera **14** is provided on the rear surface of the rearview mirror **16**. The display device **17** includes a plurality of pixels (dots).

(13) Vehicle **10** comprises an Electronic Control Unit (ECU) **20**. The cameras **14**, the display device **17**, and ECU **20** are components of the vehicle image display system **25**.

(14) As illustrated in FIG. 2, ECU **20** includes a Central Processing Unit (CPU) (processor) (computer) **20A**, Read Only Memory (ROM) **20B**, Random Access Memory (RAM) **20C**, a storage **20D**, a communication Inter Face (I/F) **20E**, an input/output I/F **20F**, and a Graphics Processing Unit (GPU) **20G**. CPU **20A**, ROM **20B**, RAM **20C**, the storage **20D**, the communication I/F **20E**, the input/output I/F **20F**, and GPU **20G** are communicably connected to each other via a bus **20Z**.

(15) CPU **20A** is a central processing unit. CPU **20A** executes various programs and controls each unit. That is, CPU **20A** reads the program from ROM **20B** or the storage **20D**, and executes the program using RAM **20C** as a working area. In accordance with the programs recorded in ROM **20B** or the storage **20D**, CPU **20A** performs control of the respective components and various arithmetic processes (information processing).

(16) ROM **20B** stores various programs and various data. RAM **20C** temporarily stores a program/data as a working area. The storage **20D** is constituted by a storage device such as Hard Disk Drive (HDD) or Solid State Drive (SSD). The storage **20D** stores various programs and various data. The ROM **20B** and the storage **20D** are examples of a storage medium. A communication I/F **20E** is an interface through which an ECU **20** communicates with other devices. The input/output T/F **20F** are interfaces for communicating with various devices. For example, the camera **14** and the display device **17** are connected to the input/output I/F **20F**.

(17) GPU **20G** is an arithmetic unit for image-processing. GPU **20G** reads a program from a ROM **20B** or a storage **20D**, and executes the program using RAM **20C** as a working area.

(18) FIG. 3 shows a functional diagram of an ECU **20**. ECU **20** includes a luminance determination unit **201**, a pixel value determination unit **202**, an averaged luminance determination unit (dark condition determination unit) **203**, a camera control unit (image adjustment unit) **204**, and a display device control unit **205** as functional components. These are realized by a CPU **20A** and a GPU **20G** reading and executing a program stored in a ROM **20B**.

(19) The luminance determination unit **201** measures the luminance of all the pixels included in the image data received from the camera **14**. In addition, the luminance determination unit **201** determines whether or not a high-luminance pixel is included in these pixels. The high-luminance pixel is a pixel having a luminance equal to or higher than the first threshold. The first threshold value is, for example, a maximum luminance value (**255**).

(20) The pixel value determination unit **202** measures pixel values of all pixels included in the image data received from the camera **14**. In addition, the pixel value determination unit **202** determines whether or not a pixel having a pixel value equal to or less than the second threshold value is included in these pixels. When a pixel having a pixel value equal to or less than the second threshold value is displayed on the display device **17**, the occupant who views the display device **17** recognizes the image represented by the pixel as black.

(21) The average luminance determination unit **203** measures the average luminance of all the pixels included in the image data received from the camera **14**. The average luminance determination unit **203** determines whether or not the average luminance is equal to or less than the third threshold value.

(22) The camera control unit **204** has a brightness adjustment function. That is, the camera control unit **204** controls the camera **14**. Thus, the camera control unit **204** can execute the brightness adjustment process. The brightness adjustment processing is processing for adjusting the brightness of all the pixels included in the image data. The brightness adjustment processing includes a first

brightness adjustment processing, a second brightness adjustment processing, and a third brightness adjustment processing, which will be described later. The camera control unit **204** adjusts the brightness of the entire image data (all pixels) based on the average luminance values of all the pixels included in the image data. More specifically, the camera control unit **204** executes brightness adjustment processing by at least one of exposure adjustment and gain adjustment. The camera control unit **204** calculates an exposure time based on the average luminance value. The camera control unit **204** controls an exposure adjustment system (electronic shutter) of the camera **14** based on the calculated exposure time information. The camera control unit **204** adjusts an amplification factor (gain) of an electric signal output from all the pixels (dots) of the image sensor based on the average luminance value.

(23) The display device control unit **205** outputs the image data whose brightness has been adjusted to the display device **17**. The display device control unit **205** causes the display device **17** to display a digital image based on the image data.

(24) Next, the flow of the process performed by ECU **20** will be described with reference to the flow chart of FIG. **4**. It is assumed that, as shown in FIG. **1**, the vehicle **10** and the following vehicle **30** travel forward on the road **40**, and the lighting device **31** of the following vehicle **30** emits the illumination light **32** toward the front. Further, as shown in FIGS. **5** and **6**, the camera **14** photographs the following vehicle **30**. It is assumed that a digital image representing the following vehicle **30** is displayed on the display device **17** of the rearview mirror **16**. Note that reference numeral **41** in FIG. **5** and FIG. **6** is a white line indicating both side edges of the road **40**.

(25) First, in S**10** (hereinafter, the characters of “steps” are omitted), ECU **20** (average luminance determination unit **203**) measures the average luminance of all the pixels included in the image data received from the camera **14**. Further, ECU **20** determines whether the mean brightness is less than or equal to the third threshold.

(26) When Yes is determined by S**10**, it is highly likely that the surrounding environment of the vehicles **10** is a dark environment. For example, it is highly likely that the vehicle **10** is traveling on the road **40** at night. When Yes is determined in S**10**, a predetermined dark condition is satisfied. On the other hand, when it is determined that S**10** is No, it is highly likely that the surrounding environment of the vehicle **10** is a bright environment. For example, it is highly likely that the vehicle **10** is traveling on the road **40** in the daytime.

(27) When Yes is determined in S**10**, ECU **20** proceeds to S**11**. In S**11**, ECU **20** determines whether or not all the pixels included in the image data received from the camera **14** include high-luminance pixels. The high-luminance pixel is a pixel having a luminance equal to or higher than the first threshold.

(28) FIG. **5** shows the display device **17** when the inter-vehicle distance between the vehicle **10** and the following vehicle **30** is short. At this time, a large amount of light flux included in the illumination light **32** emitted from the lighting device **31** of the following vehicle **30** is received by the camera **14** (imaging element). That is, strong light enters the camera **14** (image sensor). In such cases, ECU **20** determines that the high-brightness pixels are included in the images received from the cameras **14**. Pixels (pixels) of the image data representing at least a part of the lighting device **31** are high-luminance pixels.

(29) On the other hand, FIG. **6** shows the display device **17** when the distance between the vehicle **10** and the following vehicle **30** is long. At this time, a small amount of light flux included in the illumination light **32** emitted from the lighting device **31** of the following vehicle **30** is received by the camera **14** (imaging element). That is, no strong light enters the camera **14** (image sensor). In such cases, ECU **20** determines that the high-brightness pixels are not included in the images received from the cameras **14**.

(30) When Yes is determined in S**11**, ECU **20** proceeds to S**12**. ECU **20** measures pixel values of all pixels included in the image data received from the camera **14**. In addition, ECU **20** determines whether or not a pixel having a pixel value equal to or less than the second threshold value is

included in a peripheral pixel that is a pixel around the high-luminance pixel. The range of the peripheral pixel may be, for example, a predetermined multiple of the range of the high-luminance pixel. The range of the peripheral pixels may be a range in which the entire object having the light source (the following vehicle **30** in the present embodiment) is included.

(31) When it is determined that **S12** is No, a predetermined process condition is satisfied. When **S12** is determined to be No and the first brightness adjusting process is not performed, the digital images displayed on the display device **17** are highly likely to have so-called white blur. The image data (pixels) generated by a specific pixel included in the pixel array portion of the image sensor and being a part of the pixels (dots) that have received strong light is a high-luminance pixel. In this case, light that cannot be absorbed by a specific pixel (dot) is absorbed by a peripheral pixel that is a pixel (dot) around the specific pixel. Therefore, when the first brightness adjustment process is not performed, the occupant recognizes that the digital image based on the peripheral pixels generated by the peripheral pixels (dots) of the image sensor is blurred.

(32) On the other hand, if a pixel having a pixel value equal to or less than the second threshold value is included in the peripheral pixel of the image data, ECU **20** determines that the pixel value is Yes in **S12**. In this case, it is unlikely that white blur has occurred in the digital image displayed on the display device **17**.

(33) If **S12** determines No, ECU **20** proceeds to **S13**. ECU **20** performs a first brightness adjusting process of reducing brightness of all pixels of the image data. That is, ECU **20** performs at least one of the exposure adjustment and the gain adjustment based on the mean brightness of all the pixels acquired by **S10**.

(34) The brightness of the pixel of the image data prior to the brightness adjustment process being executed is referred to as the initial-brightness **BR1**. The brightness after the brightness adjustment process is executed is referred to as a corrected brightness **BR2**. Furthermore, the difference between the initial brightness **BR1** and the corrected brightness **BR2** is referred to as a brightness reduction amount **BRA** (=initial brightness **BR1**−corrected brightness **BR2**).

(35) Also, when No is determined by **S11** or Yes is determined by **S12**, ECU **20** proceeds to **S14**. ECU **20** performs a second brightness adjusting process based on the mean brightness of all the pixels acquired by **S10**. The second brightness adjustment process is a process of adjusting the brightness of all the pixels of the image data. The second brightness adjustment process is a process of adjusting the brightness of the image data so that the occupant of the vehicle **10** can easily visually recognize the digital image displayed on the display device **17** when the digital image displayed on the display device **17** is unlikely to be blurred and the surrounding environment of the vehicle **10** is a dark environment. Note that the brightness reduction amount **BRA** in the case where the brightness of the image-data is reduced by the second brightness adjustment processing is smaller than in the case where the first brightness adjustment processing is executed. That is, by the second brightness adjustment processing, for example, the brightness of the image data on which the digital image representing the white line **41** shown in FIG. **6** is based is not unnecessarily reduced. Therefore, the occupant who views the display device **17** shown in FIG. **6** can clearly view the white line **41**.

(36) If **S10** determines No, ECU **20** proceeds to **S15**. ECU **20** performs a third brightness adjustment process of adjusting the brightness of all the pixels of the image data based on the mean brightness of all the pixels acquired by **S10**. The third brightness adjustment processing is processing for adjusting the brightness of the image data so that the occupant of the vehicle **10** can easily visually recognize the digital image displayed on the display device **17** when the surrounding environment of the vehicle **10** is a bright environment. Therefore, in the third brightness adjustment processing, the brightness of the image data is not reduced.

(37) When **S15** has been processed from **S13**, ECU **20** proceeds to **S16**. ECU **20** transmits the image-data subjected to the brightness adjusting process to the display device **17**. Therefore, the display device **17** displays a digital image based on the image data subjected to the brightness

adjustment processing.

(38) When the process of **S16** is completed, **ECU 20** temporarily ends the process of the flow chart of **FIG. 4**.

(39) As described above, when the dark condition is satisfied, the image data includes a high-luminance pixel that is a pixel having a luminance equal to or higher than the first threshold value, and the peripheral pixel does not include a pixel having a pixel value equal to or lower than the second threshold value, the system **25** of the present embodiment determines that the predetermined processing condition is satisfied. That is, based on whether or not pixels having pixel values equal to or less than the second threshold value are included in the image data, the system **25** determines whether or not the digital image displayed on the display device **17** is highly likely to be blurred. When the determination is made based on the pixel value as described above, it is determined with high accuracy whether or not the digital image displayed on the display device **17** is highly likely to be blurred. Further, when the processing condition is satisfied, the system **25** executes the first brightness adjustment processing for suppressing the occurrence of the white blur with respect to the image data. Therefore, it is difficult for a person who views the display device **17** displaying the digital image based on the image data subjected to the first brightness adjustment processing to feel glare.

(40) On the other hand, if the dark condition is not satisfied, if the image data does not include a high-luminance pixel, or if the peripheral pixel includes a pixel having a pixel value equal to or smaller than the second threshold value, it is considered that the possibility of occurrence of white blur in the peripheral pixel is low. Therefore, the system **25** does not perform the first brightness adjustment process on the image data in this case. That is, when the processing condition is not satisfied, the system **25** does not unnecessarily reduce the brightness of the image data.

(41) Further, the system **25** executes a second brightness adjustment process different from the first brightness adjustment process in a case where the high-luminance pixel is not included in the image data or in a case where the peripheral pixel includes a pixel having a pixel value equal to or smaller than the second threshold. That is, in this case, the system **25** adjusts the brightness of the image data so that the occupant of the vehicle **10** can easily visually recognize the digital image displayed on the display device **17**.

(42) Further, the system **25** determines that the dark condition is satisfied when the average luminance of the image data is equal to or less than the third threshold value. Therefore, the system **25** can accurately determine whether or not the surrounding environment of the vehicle **10** is dark. Therefore, the system **25** can accurately determine whether or not there is a risk of occurrence of white blur in the display device **17**.

(43) The vehicle image display system, the vehicle image display method, and the program according to the embodiment have been described above. However, the vehicle image display system, the vehicle image display method, and the program can be appropriately changed in design without departing from the gist of the present disclosure.

(44) For example, the vehicle **10** may include an illuminance detection device that detects illuminance around the vehicle **10**. When the illuminance detected by the illuminance detecting device is equal to or less than the fourth threshold, **ECU 20** may determine that the dark condition is satisfied.

(45) The camera **14** may include a variable diaphragm device facing the image sensor. In this case, **ECU 20** may perform the brightness adjustment process by adjusting the aperture diameter of the variable aperture device.

(46) **ECU 20** may perform the brightness adjusting process by performing the image processing on the image data.

(47) The display device **17** may be a display device provided on an instrument panel of the vehicle **10**.

(48) The camera **14** may be a camera capable of capturing an image of a subject located in a

direction different from the rear of the vehicle **10**.

(49) The vehicle **10** may include a detection device capable of detecting another vehicle located in the vicinity of the vehicle **10** and detecting a distance between the vehicle **10** and another vehicle. When no other vehicle is detected, or when the distance between the vehicle **10** and the other vehicle is equal to or greater than a predetermined value, ECU **20** may determine that the possibility of the occurrence of the white blur is low.

(50) Additional Remarks

(51) The vehicle image display system of the present disclosure may be any combination of the following configurations 1 to 5.

(52) <Configuration 1> A vehicle image display system includes an imaging device, a display device, a dark condition determination unit, a luminance determination unit, a pixel value determination unit, and an image adjustment unit. The imaging device images a subject located around a vehicle. The display device displays a digital image that is an image based on image data generated by the imaging device. The dark condition determination unit determines whether a dark condition is satisfied with respect to a brightness of a surrounding environment of the vehicle. The luminance determination unit determines whether a high luminance pixel that is a pixel of a luminance equal to or larger than a first threshold value is included in the image data. The pixel value determination unit determines whether a pixel having a pixel value equal to or smaller than a second threshold value is not included in a peripheral pixel that is a pixel around the high luminance pixel. The image adjustment unit performs a first brightness adjustment process for reducing a brightness of the image data, when the dark condition is satisfied and a processing condition is satisfied, the processing condition being a condition that is satisfied when the image data includes the high luminance pixel and the peripheral pixel does not include a pixel having a pixel value equal to or smaller than the second threshold value.

(53) <Configuration 2> In the vehicle image display system, when the dark condition is satisfied and the processing condition is not satisfied, the image adjustment unit performs a second brightness adjustment process different from the first brightness adjustment process on the image data.

(54) <Configuration 3> In the vehicle image display system, the dark condition is satisfied when an average luminance of the image data is equal to or smaller than a third threshold value or when an illuminance around the vehicle is equal to or smaller than a fourth threshold value. Further, the vehicle image display method of the present disclosure may be a combination of the following configuration 4 and at least one of the configurations 1 to 3.

(55) <Configuration 4> The vehicle image display method includes: a step of determining whether a dark condition is satisfied with respect to a brightness of a surrounding environment of a vehicle; a step of determining whether a high luminance pixel that is a pixel of a luminance equal to or larger than a first threshold value is included in an image data generated by an imaging device that images a subject located around the vehicle; a step of determining whether a pixel having a pixel value equal to or smaller than a second threshold value is not included in a peripheral pixel that is a pixel around the high luminance pixel; a step of performing a first brightness adjustment process for reducing a brightness of the image data, when the dark condition is satisfied and a processing condition is satisfied, the processing condition being a condition that is satisfied when the image data includes the high luminance pixel and the peripheral pixel does not include a pixel having a pixel value equal to or smaller than the second threshold value; and a step of displaying a digital image that is an image based on the image data subjected to the first brightness adjustment process. Further, the program of the present disclosure may be a combination of the following configuration 5 and at least one of the configurations 1 to 3.

(56) <Configuration 5> The program causes a computer to execute processes including: a process of determining whether a dark condition is satisfied with respect to a brightness of a surrounding environment of a vehicle; a process of determining whether a high luminance pixel that is a pixel of

a luminance equal to or larger than a first threshold value is included in an image data generated by an imaging device that images a subject located around the vehicle; a process of determining whether a pixel having a pixel value equal to or smaller than a second threshold value is not included in a peripheral pixel that is a pixel around the high luminance pixel; a process of performing a first brightness adjustment process for reducing a brightness of the image data, when the dark condition is satisfied and a processing condition is satisfied, the processing condition being a condition that is satisfied when the image data includes the high luminance pixel and the peripheral pixel does not include a pixel having a pixel value equal to or smaller than the second threshold value; and a process of displaying a digital image that is an image based on the image data subjected to the first brightness adjustment process.

Claims

1. A vehicle image display system comprising: an imaging device that images a subject located around a vehicle; a display device that displays a digital image that is an image based on image data generated by the imaging device; a dark condition determination unit that determines whether a dark condition is satisfied with respect to a brightness of a surrounding environment of the vehicle; a luminance determination unit that determines whether a high luminance pixel that is a pixel of a luminance equal to or larger than a first threshold value is included in the image data; a pixel value determination unit that determines whether a pixel having a pixel value equal to or smaller than a second threshold value is not included in a peripheral pixel that is a pixel around the high luminance pixel; and an image adjustment unit that performs a first brightness adjustment process for reducing a brightness of the image data, when the dark condition is satisfied and a processing condition is satisfied, the processing condition being a condition that is satisfied when the image data includes the high luminance pixel and the peripheral pixel does not include a pixel having a pixel value equal to or smaller than the second threshold value.
2. The vehicle image display system according to claim 1, wherein when the dark condition is satisfied and the processing condition is not satisfied, the image adjustment unit performs a second brightness adjustment process different from the first brightness adjustment process on the image data.
3. The vehicle image display system according to claim 1, wherein the dark condition is satisfied when an average luminance of the image data is equal to or smaller than a third threshold value or when an illuminance around the vehicle is equal to or smaller than a fourth threshold value.
4. A vehicle image display method comprising: a step of determining whether a dark condition is satisfied with respect to a brightness of a surrounding environment of a vehicle; a step of determining whether a high luminance pixel that is a pixel of a luminance equal to or larger than a first threshold value is included in an image data generated by an imaging device that images a subject located around the vehicle; a step of determining whether a pixel having a pixel value equal to or smaller than a second threshold value is not included in a peripheral pixel that is a pixel around the high luminance pixel; a step of performing a first brightness adjustment process for reducing a brightness of the image data, when the dark condition is satisfied, the image data includes the high luminance pixel, and the peripheral pixel does not include a pixel having a pixel value equal to or smaller than the second threshold value; and a step of displaying a digital image that is an image based on the image data subjected to the first brightness adjustment process.
5. A non-transitory storage medium storing a program that causes a computer to execute processes comprising: a process of determining whether a dark condition is satisfied with respect to a brightness of a surrounding environment of a vehicle; a process of determining whether a high luminance pixel that is a pixel of a luminance equal to or larger than a first threshold value is included in an image data generated by an imaging device that images a subject located around the vehicle; a process of determining whether a pixel having a pixel value equal to or smaller than a

second threshold value is not included in a peripheral pixel that is a pixel around the high luminance pixel; a process of performing a first brightness adjustment process for reducing a brightness of the image data, when the dark condition is satisfied, the image data includes the high luminance pixel, and the peripheral pixel does not include a pixel having a pixel value equal to or smaller than the second threshold value; and a process of displaying a digital image that is an image based on the image data subjected to the first brightness adjustment process.
